



## WeMOS D1 R32 ESP32 Wi-Fi and Bluetooth Board

This is the Wemos D1 R32, an ESP32 development board in Arduino Uno form factor. It is running on clock speed up to 240MHz and one core is dedicated to wireless functions, another core reserved for user application. This is much faster than its predecessor, ESP8266. Although it is a 3.3V system, with it in Arduino UNO form factor, you should be able to utilize most of the shields out there.



**SKU: [MDU1143](#)**

### **Brief Data:**

- Form Factor: Arduino UNO and headers.
- CPU and Memory: Xtensa® 32-bit LX6 Dual-core processor, up to 600 DMIPS.
- Operating Voltage: (5~12)Vdc.
- I/O pins Voltage: 3.3V.
- 4-MByte SPI Flash, 448-KByte ROM, 520-KByte SRAM.
- WiFi: 802.11 b/g/n/e/i.
- Bluetooth V4.2 BR/EDR and BLE specification.
- Flash: 4-Mbytes.
- Interface: Micro USB connection.
- USB Bridge IC: CH340G.
- Board Dimension: (68x53x3.1) mm.

## Mechanical Dimension:

Unit: mm



## **Tutorial:**

This is a short tutorial on how to have a simple ESP32 project running with Arduino IDE.

In order to start this tutorial, we need the following:

1. Wemos R32 development board.
2. Arduino IDE.

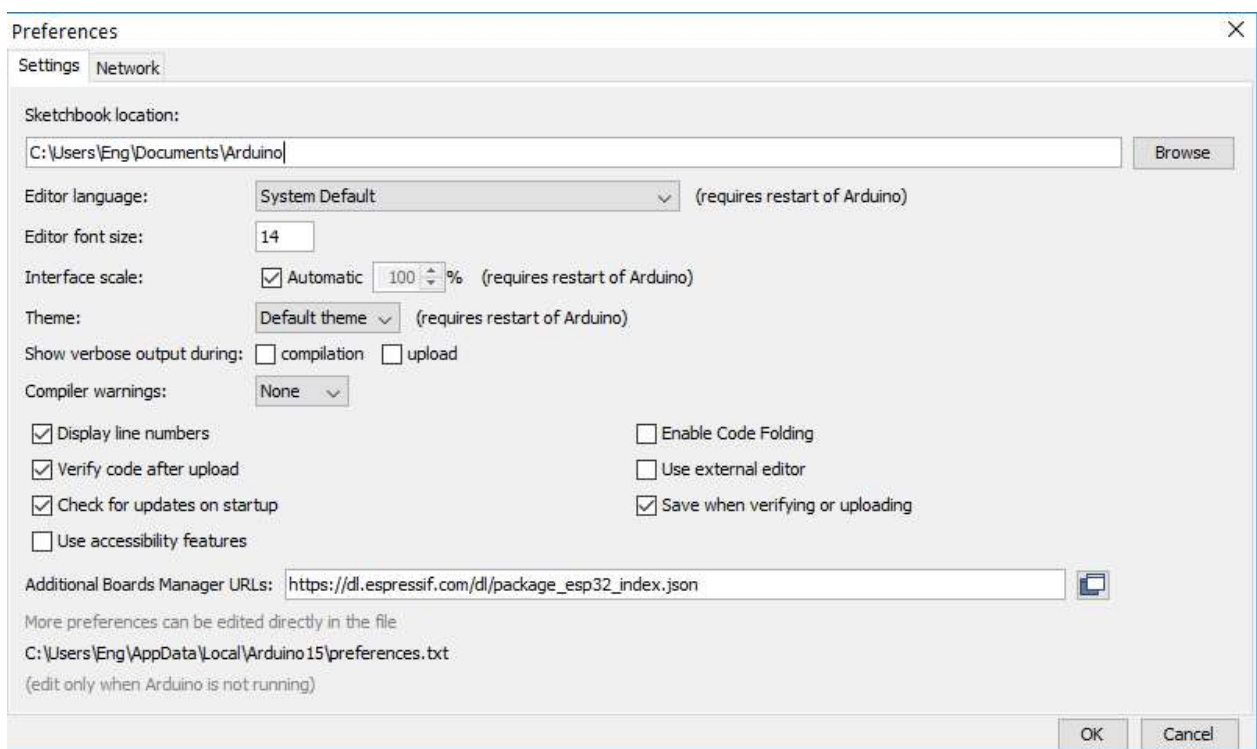
Procedure:

1. Before we plug into the Wemos R32 board, we need to install Arduino IDE first. Get the Arduino from this link: <https://www.arduino.cc/en/Guide/HomePage>
2. Afterwards, open the Arduino and obtain the ESP32 support first:

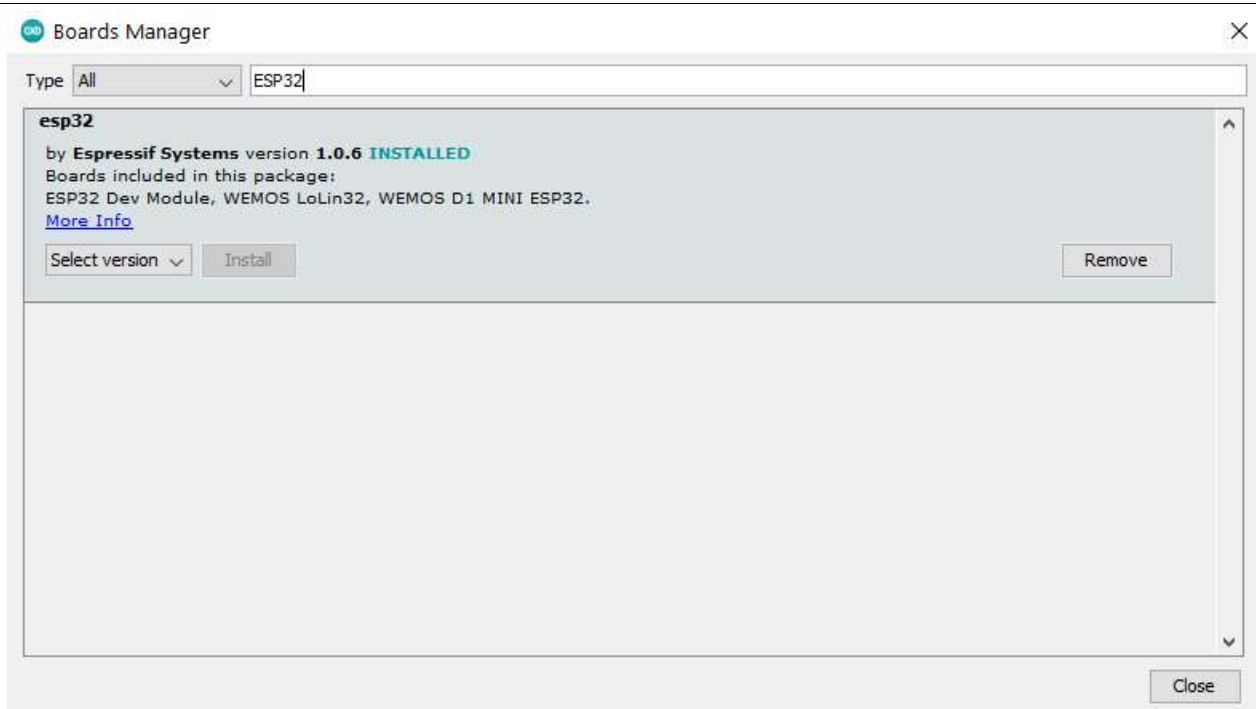
Select **File>Preferences**, and then copy this link:

[https://dl.espressif.com/dl/package\\_esp32\\_index.json](https://dl.espressif.com/dl/package_esp32_index.json) into the

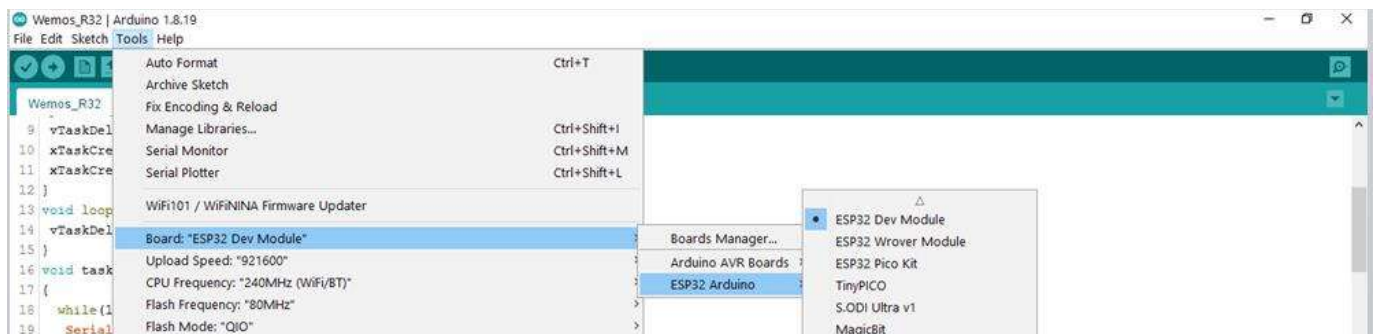
“Additional Boards Manager URLs” and click OK.



3. Go to **Tools>Board>Boards Manager**, then type in "ESP32". When you see this, press **Install**:



4. Once it's all done, go to **Tools>Board>Boards Manager**, and then select **"ESP32 Dev Module"**:



5. Create a new project! Type in the sketch below:

```
void setup() {
    Serial.begin(112500);

    pinMode(2, OUTPUT);

    vTaskDelay(1000 / portTICK_PERIOD_MS);

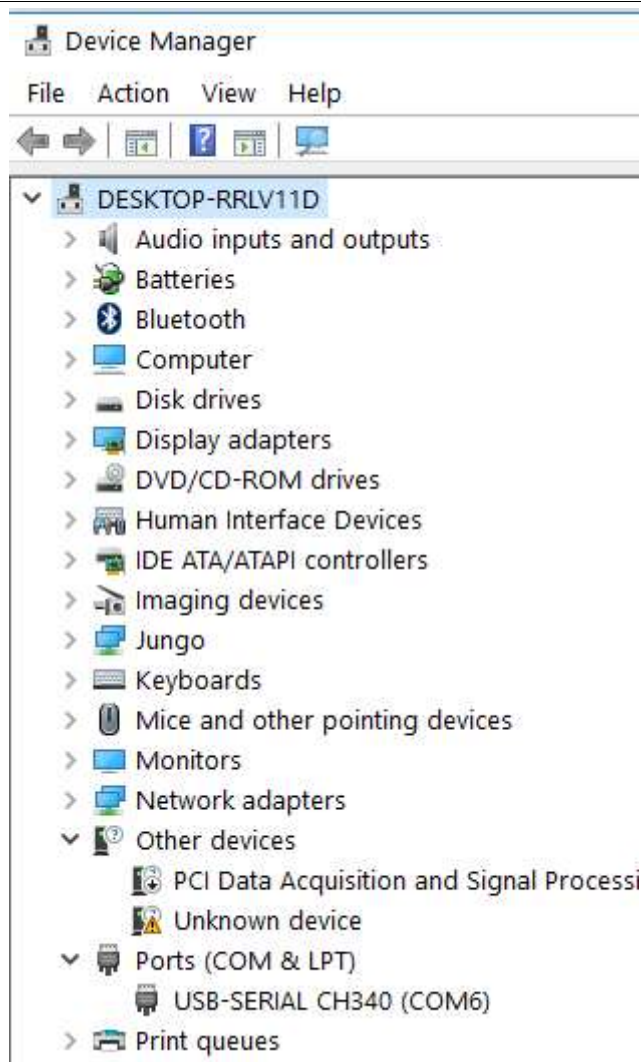
    xTaskCreate(task1, "task1", 2048, NULL, 1, NULL);

    xTaskCreate(task2, "task2", 2048, NULL, 1, NULL);
}

void loop() {
    vTaskDelay(1000 / portTICK_PERIOD_MS);
}
```

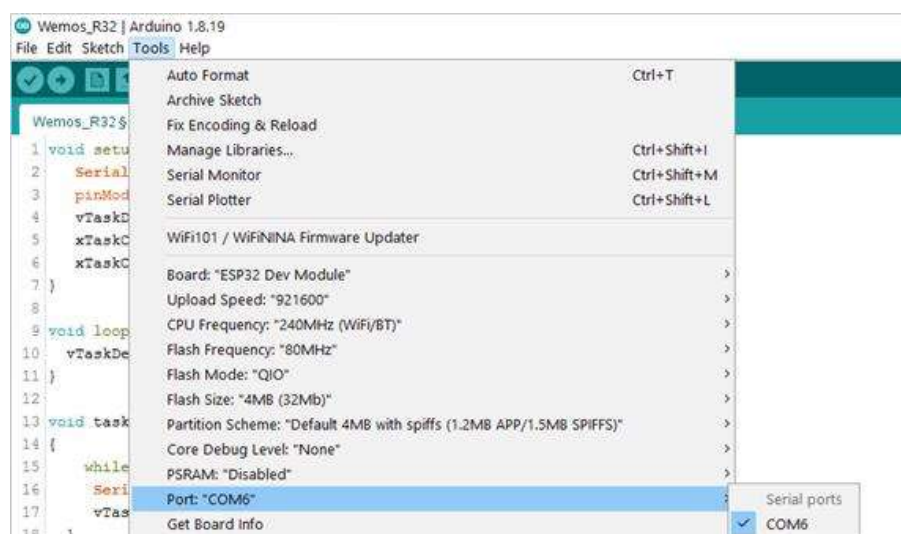
```
void task1( void * parameter )  
  
{  
  
    while(1) {  
  
        Serial.println("Hello Handsontec!");  
  
        vTaskDelay(1000 / portTICK_PERIOD_MS);  
  
    }  
  
}  
  
void task2( void * parameter)  
  
{  
  
    while(1) {  
  
        digitalWrite(2, HIGH);  
  
        vTaskDelay(1000 / portTICK_PERIOD_MS);  
  
        digitalWrite(2, LOW);  
  
        vTaskDelay(1000 / portTICK_PERIOD_MS);  
  
    }  
  
}
```

6. Save the project.
7. Before you upload, connect the R32 board to your PC. Find the COM port that is with the R32 board through Device Manager. The board is equipped with the CH340 USB bridge IC, so take note of the COM number:



In this example it's COM6, so it will be different on your PC. Note: If you couldn't find this "USB-SERIAL CH340", the drivers are probably not installed. You can refer to the CH340 drivers installation guide there: <https://sparks.gogo.co.nz/ch340.html>

8. Back to your Arduino IDE, select your COM port that is connected to the board:



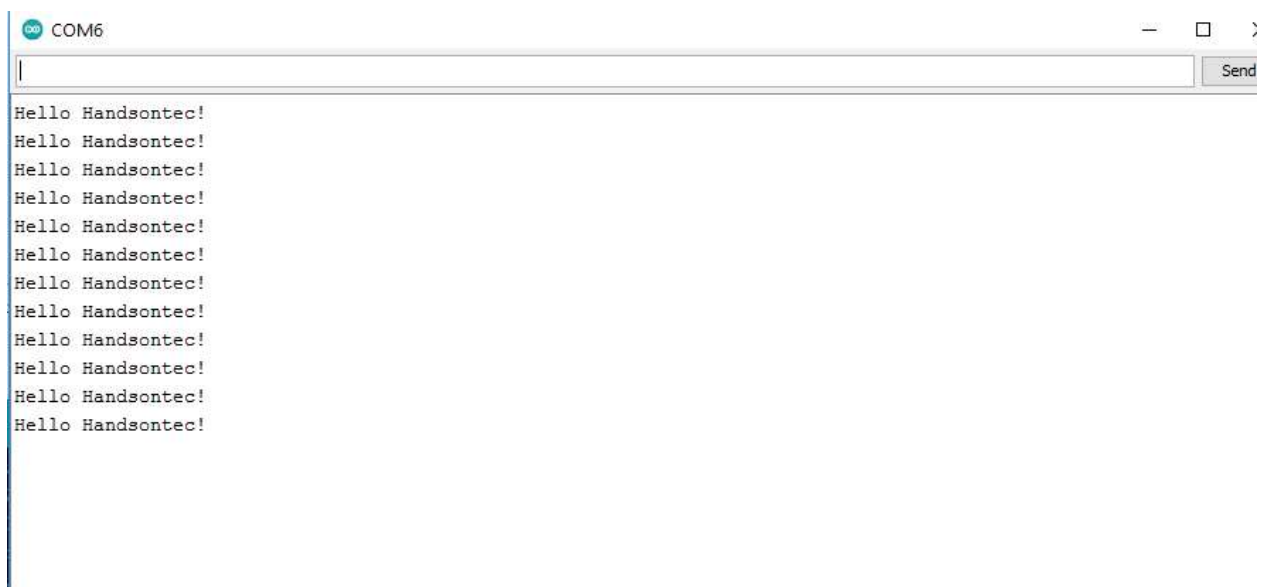
9. Press "Upload". You will see a lot of messages scrolling downwards. It is in the process of uploading the binary to the ESP32 module on the board.

```
Done uploading.
Hash of data verified.
Compressed 3072 bytes to 128...
Writing at 0x00008000... (100 %)
Wrote 3072 bytes (128 compressed) at 0x00008000 in 0.0 seconds (effective 2234.2 kbit/s)...
Hash of data verified.

Leaving...
Hard resetting via RTS pin...
Invalid library found in C:\Users\Eng\Documents\Arduino\libraries\arduino-esp8266littlefs-plugin-master: n
Invalid library found in C:\Users\Eng\Documents\Arduino\libraries\ESP8266LittleFS-2.6.0: no headers files
Invalid library found in C:\Users\Eng\Documents\Arduino\libraries\LCD1602-I2C-library-master: no headers f
Invalid library found in C:\Users\Eng\Documents\Arduino\libraries\SD-master: no headers files (.h) found i
```

The upload is successful when it says : **"Hash of data verified. Leaving..."** at the end of the process.

10. Open a terminal program and you will see "Hello World!" printed on it each second, and the LED on the board toggles at the rate of 500ms:



```
COM6
Hello Handsontec!
Hello Handsontec!
Hello Handsontec!
Hello Handsontec!
Hello Handsontec!
Hello Handsontec!
Hello Handsontec!
Hello Handsontec!
Hello Handsontec!
Hello Handsontec!
Hello Handsontec!
```

**You are done here! Congratulation, the ESP board is working.**

## **Web Resources:**

- <https://www.hackster.io/NYH-workshop/wemos-r32-with-arduino-startup-guide-7bc841>



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